ANALYSTLAB AFRICA

Experience Lab Internship Programme

Official Weekly Assignment

Week 4: FinTrust Digital Bank — Test, Refine & Present

Document Control

Week	Week 4
Project	FinTrust Financial Intelligence & Digital Banking Support Solution
Version	1.0
Issue Date	_____
Submission Deadline	End of Week 4 (Sunday, 11:59 PM WAT)

1. WEEK 4 OVERVIEW

Week 4 is the **final stage of the FinTrust Digital Bank Experience Lab**.

During Weeks 1–3, you have:

- Understood the FinTrust business problem.
- Reviewed the project resources.
- Analyzed and prepared the available data.
- Developed your track-specific solution.
- Tested and refined components of your work.
- Prepared outputs for integration.

This week, your focus is to:

Test → Validate → Refine → Finalize → Present

You are expected to complete your track-specific solution, conduct final testing and validation, address identified issues, document limitations and prepare your work for the final FinTrust solution.

The FinTrust roadmap identifies Week 4 as the stage for final testing, refinement, final integration and presentation.

2. PROJECT RESOURCES

You should continue working with the approved FinTrust resources:

- FinTrust Project Brief
- FinTrust Customer Data
- FinTrust Transaction Data
- FinTrust Data Dictionary
- FinTrust Financial Knowledge Base
- Your Week 1 work
- Your Week 2 work
- Your Week 3 work

- Outputs generated during the project

Do not introduce an unrelated dataset or create a new project for Week 4.

3. GENERAL WEEK 4 REQUIREMENTS

All interns are expected to:

A. Review Previous Work

Review your Week 1–3 work and identify what needs to be completed, corrected or improved.

B. Complete Your Track-Specific Solution

Your Week 4 work should represent the **final version** of your assigned track output.

C. Conduct Final Testing

Test your final output using appropriate technical, analytical or functional checks.

D. Refine Your Work

Address meaningful issues identified during testing.

E. Validate Results

Provide evidence that your final output works as intended and that your conclusions are supported by your work.

F. Document Limitations

Clearly state what your solution can and cannot demonstrate.

G. Prepare for Final Integration

Ensure your outputs can be clearly understood and incorporated into the wider FinTrust solution.

H. Prepare Your Individual Presentation

Every intern must prepare an individual **5–10 minute recorded presentation** covering their contribution, results, integration, limitations and lessons learned.

4. DATA ANALYTICS TRACK

Task: FinTrust Final Business Intelligence & Dashboard Solution

Your objective is to finalize your FinTrust analytical solution and demonstrate how your analysis supports business understanding and decision-making.

Part A — Final Data & Analysis Review

Review your previous analysis and confirm:

- Data quality issues have been addressed.
- Calculations are accurate.
- SQL queries produce expected results.
- Python analysis is reproducible.
- KPIs are clearly defined.
- Findings are supported by evidence.

Document any corrections or improvements made.

Part B — Final SQL Analysis

Review and refine your most important SQL analyses.

You should demonstrate appropriate use of SQL techniques such as:

- `JOIN`
- `GROUP BY`
- `CASE`
- Aggregations
- Subqueries or CTEs
- Window functions where appropriate

Identify your **most important final business queries** and explain what each reveal.

Part C — Final Power BI Dashboard

Complete and refine your Power BI dashboard.

Your dashboard should appropriately communicate areas such as:

- Customer KPIs
- Transaction KPIs
- Transaction trends
- Customer segments
- Channels
- Transaction status
- Risk-related patterns

Review:

- Visual clarity
- KPI accuracy
- Filters/slicers
- Labels
- Titles
- Consistency
- User experience

Part D — Final Business Insights

Provide at least **5 final business insights**.

Use:

Finding → Evidence → Business Meaning → Recommended Action

Avoid presenting charts without explaining what they mean for FinTrust.

Part E — Final Validation

Validate at least **5 important analytical findings**.

Use:

Finding → Validation Method → Result → Conclusion

Document any finding that changed after validation.

Recommended Tools

Excel, SQL, Python, Pandas, NumPy, Matplotlib, Seaborn, Power BI, Jupyter/Google Colab

Deliverables

- Final Power BI dashboard
- Final SQL scripts
- Final Python notebook
- KPI documentation
- Final business insights
- Recommendations
- Validation evidence
- Data quality documentation
- Final project documentation

5. DATA SCIENCE TRACK

Task: FinTrust Final Predictive Risk Model

Your objective is to complete, evaluate and document your final predictive model using the work developed during Weeks 1–3.

Part A — Final Feature Review

Review the features developed during Week 2 and Week 3.

You may:

- Retain useful features.
- Refine existing features.
- Remove weak or redundant features.
- Replace or restructure features **where supported by analysis**.
- Introduce justified feature engineering.

Every significant feature change must have a documented reason.

Part B — Final Model Evaluation

Evaluate the candidate models developed during Week 3.

Your evaluation should include appropriate metrics such as:

- Accuracy
- Precision

- Recall
- F1-score
- ROC-AUC where appropriate
- Confusion matrix

Compare the candidate models and document the reasoning behind your final model selection.

Logistic Regression should remain an important interpretable baseline; however, it does not automatically have to be the final model.

Part C — Error Analysis

Analyze:

- False positives
- False negatives
- True positives
- True negatives

Identify where the model performs well and where it struggles.

Part D — Final Model Interpretation

Explain:

- Important features
- How features influence predictions
- Model strengths
- Model weaknesses
- Important assumptions

Use appropriate interpretation techniques where applicable.

Part E — Data Analytics Comparison

Compare important Data Science findings with relevant Data Analytics findings where outputs are available.

For example:

DA Finding → DS Feature/Model Test → DS Result → Comparison → Interpretation

The results **do not need to be identical**.

If DA identifies an important pattern but DS finds that the feature does not materially improve predictive performance, document that difference rather than forcing the results to agree.

Part F — Final Model Documentation

Document:

- Final dataset/features
- Final model
- Model selection reasoning
- Evaluation results
- Error analysis
- Limitations
- Validation evidence
- Responsible-use considerations

Important: `Risk_Review_Flag` is a synthetic educational target and must not be presented as evidence of actual fraud or financial crime.

Recommended Tools

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter/Google Colab, Git/GitHub

Deliverables

- Final modelling dataset
- Final feature set
- Model comparison
- Final candidate model
- Evaluation metrics
- Confusion matrix
- Error analysis
- Model interpretation
- Validation evidence
- Limitations documentation
- Final technical report
- Updated GitHub repository

6. MACHINE LEARNING ENGINEERING TRACK

Task: FinTrust Final ML Workflow & Technical Validation

Your objective is to finalize the technical workflow developed during Weeks 2–3 and demonstrate that it operates reliably from input to output.

Part A — Final Workflow

Complete the workflow:

Input → Data Validation → Preprocessing → Feature Preparation → Model → Prediction → Output

Part B — End-to-End Testing

Test the workflow using appropriate scenarios, including:

- Valid input
- Missing values
- Unexpected categories
- Invalid data types
- Empty input
- Model loading
- Prediction generation
- Output format
- Reproducibility

Document:

Test → Expected Result → Actual Result → Pass/Fail → Action Taken

Part C — Model/Pipeline Integration

Where applicable, integrate the final model into the technical workflow.

If the final Data Science model is not available, use the agreed interface or an appropriately documented model component.

Do not claim integration that did not actually occur.

Part D — API/Service

Where appropriate, finalize your API or service component.

Demonstrate:

- Input handling
- Prediction/request processing
- Output generation
- Error handling

- Basic testing

Part E — Reproducibility

A reviewer should be able to understand:

1. How to install dependencies.
2. How to prepare the data.
3. How to run the workflow.
4. How to generate predictions.
5. How to interpret the output.

Recommended Tools

Python, Git, GitHub, Scikit-learn, VS Code/Jupyter, FastAPI where appropriate, testing tools, Docker concepts

Deliverables

- Final repository
- Final ML workflow
- Pipeline components
- Model integration
- Test results
- API/service where applicable
- `README.md`
- Requirements/dependency file
- Reproducibility evidence
- Technical documentation

7. GENERATIVE AI TRACK

Task: FinTrust Final Customer Support Assistant

Your objective is to finalize the FinTrust customer support assistant and demonstrate that it provides grounded, safe and appropriate responses.

Part A — Final Knowledge Grounding

Review the assistant's responses against the approved FinTrust Knowledge Base.

Identify and correct:

- Unsupported answers
- Hallucinations
- Missing information
- Incorrect interpretations

- Weak escalation behavior

Part B — Final Prompt Refinement

Review important prompts developed during Weeks 2–3.

Document:

Original Prompt → Problem → Revised Prompt → Expected Improvement → Actual Result

Part C — Final Grounding Evaluation

Test the assistant using:

- Supported questions
- Unsupported questions
- Ambiguous questions
- Escalation scenarios
- Knowledge-grounding tests
- Hallucination tests

Part D — Final Safety Testing

Test how the assistant responds to:

- Requests for account credentials
- Security-sensitive questions
- Personalized financial advice
- Unsupported policies
- Unsupported fees or limits
- Questions outside the approved knowledge base

The assistant must not invent FinTrust policies, fees, limits, procedures or timelines.

Part E — Final Evaluation

Create an evaluation table:

Question Expected Response Actual Response Grounded? Safe? Action

Document the improvements made after testing.

Recommended Tools

Python, Jupyter/Google Colab, VS Code, LLM/API tools, Git/GitHub, retrieval/RAG tools, embeddings/vector search where appropriate

Deliverables

- Final assistant
- Final prompt library
- Knowledge/retrieval implementation
- Grounding evaluation
- Safety evaluation
- Guardrail documentation
- Refinement evidence
- Limitations
- Final technical documentation

8. PROJECT MANAGEMENT TRACK

Task: FinTrust Final Integration & Delivery Management

Your objective is to coordinate the final stage of the FinTrust project and ensure that the required deliverables are complete and presentation-ready.

Part A — Final Deliverable Review

Create a final tracker containing:

Track	Required Output	Status Complete?	Outstanding Work
Data Analytics	Dashboard/Analysis		
Data Science	Final Model		
ML Engineering	Final Workflow		
Generative AI	Final Assistant		
PM	Final Documentation		

Part B — Final Integration Readiness

Review whether the major project outputs can be presented as one coherent solution:

Customer & Transaction Data
↓
Analytics Findings & KPIs
↓
Predictive Risk Intelligence
↓
ML Workflow/Service
↓
Knowledge-Based GenAI Support
↓
Integrated FinTrust Solution

This flow is the illustrative solution flow established in the FinTrust roadmap.

Part C — Risk & Issue Closure

Review your previous:

- Risk register
- Issue log
- Dependency register
- Decision log
- Change log

Identify which items are:

- Closed
- Still open
- Accepted
- Deferred

Part D — Final Presentation Readiness

Prepare:

- Final presentation structure
- Key project achievements
- Track-specific contributions
- Major findings
- Challenges
- Limitations
- Lessons learned

- Final recommendations

Ensure that each intern is prepared for their **individual 5–10 minute recorded presentation**.

Part E — Lessons Learned

Document at least:

- 3 things that worked well
- 3 challenges encountered
- 3 lessons learned
- Key improvements made
- Recommendations for future project work

Recommended Tools

Excel, Google Sheets, Word/Google Docs, PowerPoint/Google Slides, Trello, Notion, Jira or another appropriate project-management tool

Deliverables

- Final deliverable tracker
- Integration readiness tracker
- Updated risk register
- Issue log
- Decision/change log
- Final project timeline/status
- Presentation readiness checklist
- Lessons learned
- Final PM report

9. FINAL TESTING & VALIDATION REQUIREMENT

Every track must demonstrate the following process:

Final Output → Test → Finding → Refinement → Re-test → Final Result

Your submission should show **evidence of what was tested and what changed as a result**.

Simply stating that the work was tested is not sufficient.

10. FINAL INTEGRATION REQUIREMENT

The FinTrust solution should demonstrate how the different track outputs contribute to one overall solution.

The integration should follow the project flow:

Customer & Transaction Data
→ Analytics Findings & KPIs
→ Predictive Risk Intelligence
→ ML Workflow/Service
→ Knowledge-Based GenAI Support
→ Integrated FinTrust Solution

Where actual outputs from another track are used, document:

- Output received
- How it was used
- What changed
- Evidence of use

Do not claim collaboration or integration that did not actually occur.

11. FINAL ASSUMPTIONS, LIMITATIONS & RESPONSIBLE USE

Each intern must document the limitations of their solution.

Remember:

- FinTrust is a fictional organisation.
- The datasets are synthetic.
- `Risk_Review_Flag` is an educational target and is not actual fraud determination.
- The solution must not be presented as a production banking system.
- GenAI outputs must remain grounded in the approved Knowledge Base.
- The project must not provide personalised financial, investment or legal advice.
- Unsupported assumptions must be clearly identified.

12. WEEK 4 FINAL SUBMISSION

Each intern must submit their work **individually**.

Your final submission should contain:

Track Deliverables

All required outputs for your assigned track.

Supporting Evidence

- Testing evidence
- Validation results
- Refinements
- Screenshots where appropriate
- Analysis/code/model/dashboard evidence

Documentation

- Final report/documentation
- Assumptions
- Limitations
- Decisions
- Lessons learned

GitHub

Where applicable:

- Updated repository
- Clean project structure
- `README.md`
- Code/notebooks
- Requirements/dependencies
- Relevant documentation

13. INDIVIDUAL FINAL VIDEO PRESENTATION

Every intern must submit an **individual 5–10 minute recorded presentation**.

Your presentation should explain:

1. The FinTrust problem.
2. Your assigned track.
3. What you personally developed.
4. Tools used.
5. Key findings/results.
6. Testing and validation.
7. Improvements made.
8. How your work connects to the wider FinTrust solution.
9. Limitations.
10. Key lessons learned.

The final presentation is an **individual requirement**, even though the overall FinTrust solution contains multiple track outputs.

14. ASSESSMENT CRITERIA

Criteria	Weight
Application of Week 1–3 Foundation	10%
Final Track-Specific Quality	25%
Testing, Validation & Refinement	15%
Final Technical/Analytical Quality	15%
Final Integration & Solution Coherence	15%
Problem-Solving & Decision-Making	5%
Documentation & Portfolio Quality	5%
Individual Final Video Presentation	10%
Total	100%

Professional Development Requirement

As part of building your professional portfolio:

- Update your GitHub repository with your Week 1 project.
- Publish a LinkedIn post describing your project
- Publish an X (Twitter) post summarizing your project.
- Tag **AnalystLab Africa** on LinkedIn.
- Tag **@analystlabafric** on X.
- Include the hashtag **#AnalystLabAfrica** on both platforms.

Professional Branding Reminder

AnalystLab Africa encourages every intern to document their learning journey professionally on LinkedIn and X. Consistently sharing your work helps build your portfolio and increases your visibility to recruiters and employers.